



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



FAA and TSA Aviation Cybersecurity Roadmap Gap Brief

Threat Report

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TLP:CLEAR | Lost Boys Cyber | FAA and TSA Aviation Cybersecurity Roadmap Gap Brief - Threat Report

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FAA and TSA Aviation Cybersecurity Roadmap Gap Brief

1. Executive Summary

Industrial Cyber reported GAO findings warning that an outdated TSA cyber roadmap and FAA implementation gaps could hinder aviation cybersecurity.

Governance gaps become operational risk when cyber accountability, budget visibility, and zero-trust implementation lag behind interconnected airport, airline, and federal aviation systems. This report converts current public reporting into a defender-facing action package with detection ideas, hunt hypotheses, and sector-specific operational considerations.

2. Intelligence Requirements

Question	Current Answer	Confidence
What activity should defenders prioritize?	FAA and TSA Aviation Cybersecurity Roadmap Gap Brief	Medium
Which environments are most exposed?	Identity-heavy, internet-facing, supplier-connected, or operationally critical environments.	Medium
What should be done today?	Validate exposure, preserve logs, run the included hunts, and brief owners on likely operational impacts.	High

3. Source Review & Confidence

Source	Contribution	Confidence	
1	Industrial Cyber - GAO warns outdated TSA cyber roadmap and FAA implementation gaps	Sector-specific governance reporting	Medium
2	U.S. GAO Reports and Testimonies	Authoritative oversight source	High
3	TSA Cybersecurity Toolkit	Aviation/transportation cybersecurity context	High

The source set includes current primary/vendor or reputable reporting where available. Claims about victims, actors, and tooling are bounded to cited sources; this report does not assert unverified attribution.

4. Threat Activity Overview

The activity is important because it affects common enterprise control points: identity, remote access, software supply chain, cloud-native trust, mobile devices, or sector-specific operations. Defenders should treat the item as a prompt for exposure validation rather than as a standalone news summary.

5. Affected Sectors and Exposure

Exposure Pattern	Why It Matters
Identity and SaaS workflows	Social engineering, token theft, and delegated access can bypass perimeter-focused controls.
Third-party or supplier dependencies	Incidents can cascade into operators through shared platforms or

	managed services.
High-availability operational services	Ransomware, fraud, or identity misuse can produce downtime, safety-adjacent disruption, or executive risk.

6. Aviation and Aerospace Sector Impact

Sector Segment	Operational Relevance	Defender Action
Airlines and airports	Shared identity, check-in, baggage, passenger-service, and vendor platforms can produce operational disruption even when aircraft safety systems are not directly affected.	Validate identity hardening, third-party access, incident communications, and fallback procedures.
Aerospace / defense manufacturers	CUI, design data, supplier portals, and engineering collaboration platforms create high-value espionage and extortion targets.	Review CMMC/NIST controls, supplier access, data-loss monitoring, and segmented engineering environments.
Satellite / space / avionics supply chain	Dependencies on specialized vendors and remote operations increase third-party and identity risk.	Inventory exposed portals, privileged vendor accounts, and remote maintenance paths.

Likely exposure paths include compromised SaaS identities, supplier-managed applications, remote support tooling, public-facing portals, and unmanaged operational-technology/building-management devices at airports, MRO sites, or aerospace plants.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	Phishing, exposed service abuse, supplier compromise, or credential misuse	Consistent with current reporting and common intrusion paths for this topic.
Execution / Persistence	Scripted tooling, cloud/API misuse, or malware execution depending on environment	Validate with endpoint, identity, and application logs.
Impact	Data theft, fraud, operational disruption, ransomware, or trust-boundary abuse	Prioritize business-critical workflows and third-party access paths.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
Cited source URLs and product names	Reference observable	Seed exposure review and intelligence watchlists.
New administrative sessions or OAuth grants	Behavioral observable	Identify account takeover or delegated-access abuse.
Unusual archive/encryption/network-transfer utilities	Behavioral observable	Detect staging, exfiltration, or ransomware preparation.

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
New external sign-in followed by privileged action	Entra ID / IdP / SaaS audit logs	Find account takeover or social-engineering success.
Suspicious script or archive execution	EDR process telemetry	Surface staging, loader, or ransomware precursor behavior.
High-volume outbound transfer or rare destination	Network telemetry / proxy	Identify exfiltration or command-and-control.

9.2. Sigma / Detection Content

```

title: Suspicious Daily Threat Activity - Identity And Scripted Follow-On
id: lbc-daily-threat-faa-and-tsa-aviation-cybersecurity-r
status: experimental
logsource:
  product: windows
  category: process_creation
detection:
  selection_tools:
    Image|endswith:
      - '\powershell.exe'
      - '\wscript.exe'
      - '\cscript.exe'
      - '\rundll32.exe'
      - '\7z.exe'
      - '\rar.exe'
  selection_suspicious_args:
    CommandLine|contains:
      - 'http'
      - 'token'
      - 'password'
      - 'invoice'
      - 'backup'
      - 'encrypt'
  condition: selection_tools and selection_suspicious_args
level: medium

```

9.3. Hunt Query

```

DeviceProcessEvents
| where Timestamp > ago(14d)
| where FileName in~ ("powershell.exe", "wscript.exe", "cscript.exe", "rundll32.exe", "7z.exe", "rar.exe")
| where ProcessCommandLine has_any ("http", "token", "password", "invoice", "backup", "encrypt")
| project Timestamp, DeviceName, AccountName, FileName, ProcessCommandLine, InitiatingProcessFileName

```

10. Threat Hunting

10.1. Hunt Hypothesis

If this activity is present, defenders will see abnormal identity events followed by automation, data staging, or external communications inconsistent with the user's normal role.

10.2. Hunt Query

```

SigninLogs
| where TimeGenerated > ago(14d)
| where ResultType == 0
| summarize Countries=dcount(Location), Apps=dcount(AppDisplayName), FirstSeen=min(TimeGenerated),

```

LastSeen=max(TimeGenerated) by UserPrincipalName, IPAddress where Countries > 1 or Apps > 5
DeviceNetworkEvents where Timestamp > ago(14d) where InitiatingProcessFileName in~ ("powershell.exe", "python.exe", "curl.exe", "wget.exe", "rundll32.exe") summarize Connections=count(), FirstSeen=min(Timestamp), LastSeen=max(Timestamp) by DeviceName, InitiatingProcessAccountName, RemoteUrl, RemoteIP where Connections > 10

11. Risk Assessment

Risk Dimension	Assessment
Likelihood	Medium to High for exposed or identity-heavy environments.
Impact	Medium to High where operational continuity, regulated data, CUI, payment workflows, or safety-adjacent services are involved.
Confidence	Medium; based on public reporting and defender analysis rather than private telemetry.

12. Recommended Actions

Priority	Owner	Action
Critical	SOC / Identity	Review recent privileged sign-ins, impossible travel, new OAuth grants, and MFA reset events.
High	Vulnerability / Platform Owners	Validate affected products, services, third-party platforms, and supplier access paths.
High	IR Lead	Prepare containment playbooks for account takeover, ransomware precursor behavior, and vendor-service disruption.
Medium	Executives / Risk	Brief business owners on operational dependency and third-party continuity risk.

13. References

- [1] Industrial Cyber - GAO warns outdated TSA cyber roadmap and FAA implementation gaps — <https://industrialcyber.co/transport/gao-warns-outdated-tsa-cyber-roadmap-and-faa-implementation-gaps-could-hinder-aviation-cybersecurity/>
- [2] U.S. GAO Reports and Testimonies — <https://www.gao.gov/reports-testimonies>
- [3] TSA Cybersecurity Toolkit — <https://www.tsa.gov/for-industry/cybersecurity-toolkit>